



برمجة الشبكات

10:00-11:00

8/4/2025

د/ علي حسين

Faculty of Computers & Information, Assiut University

Course code: IT432

4th Level

Midterm Exam

Duration: One hour

This exam for the following program(s) : General IT+CS

Points: -/20

1

* الإسم الرباعي (بالعربي فقط)

2

* رقم الجلوس

ج٣

د٣

ه٣

أ٤

ب٤

ج٤

د٤

ه٤

6

رقم الكمبيوتر * 

7

الكود * (قد تمت مراجعة بيانات الطالب ورقم الجلوس) 

8

Which of the following are not a benefit of network: 

Sharing resources

(Choose all that apply)

Load Utilization

High reliability

Cost reduction

Increase security

9

In this code

What is the purpose of using `SocketType.Raw` and `ProtocolType.Icmp` in the code?



```
1  using System;
2  using System.Net;
3  using System.Net.Sockets;
4
5  class Exam
6  {
7      public static void Main()
8      {
9          string targetIp = "8.8.8.8";
10         Socket pingSocket = new Socket(AddressFamily.InterNetwork, SocketType
11         IPAddress ipAddress = IPAddress.Parse(targetIp);
12         IPEndPoint remoteEndPoint = new IPEndPoint(ipAddress, 0);
13         Console.WriteLine("Socket created for ICMP ping to " + targetIp);
14     }
15 }
16
```

`SocketType.Raw` allows the socket to send and receive custom TCP packets, while `ProtocolType.Icmp` is used for handling HTTP requests.

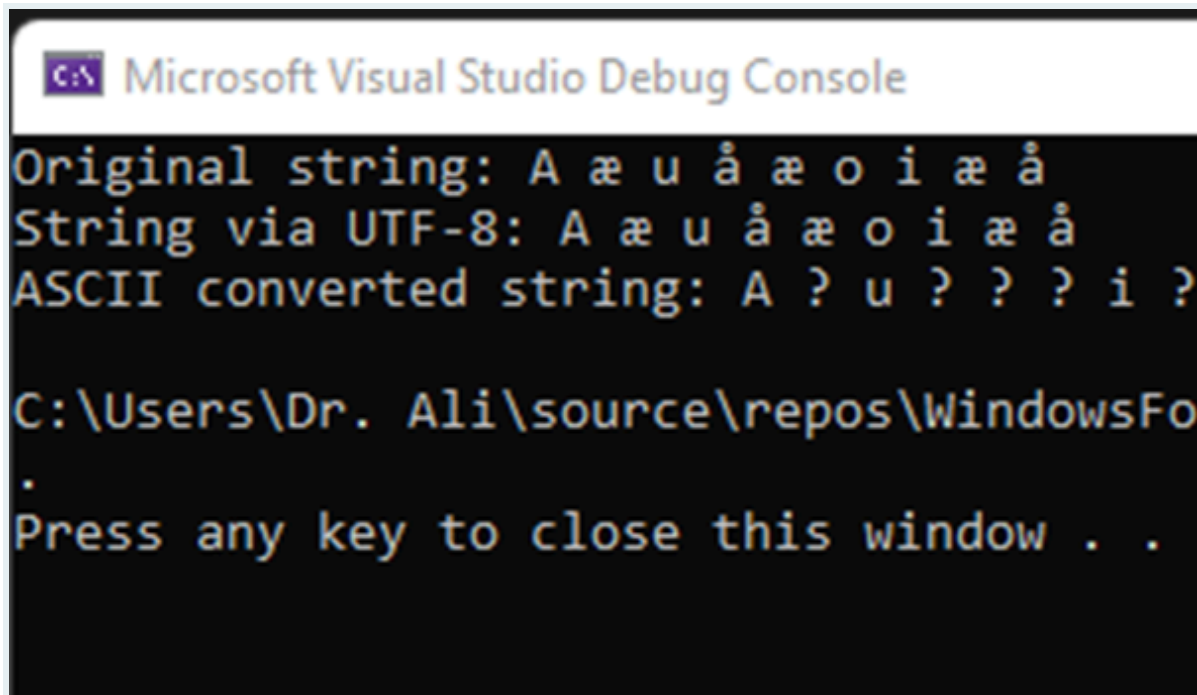
`SocketType.Raw` is used to send raw IP packets, and `ProtocolType.Icmp` is used to send and receive ICMP Echo Request and Echo Reply messages.

SocketType.Raw allows for using UDP protocols, while ProtocolType.Icmp is a placeholder for other protocols.

SocketType.Raw is used for encrypted communication, while ProtocolType.Icmp is for establishing a secure connection.

10

Identify the problem that led to the appearance of question marks
(Choose all that apply)



The ASCII encoding can only represent characters from the basic Latin alphabet, and characters like "æ", "å", and "ø" are outside its representable range.

The question marks are a result of improper conversion between encodings, where the original non-ASCII characters are lost during conversion.

The client application does not support Unicode, so all special characters are replaced with question marks.

The question marks appear because the server used an invalid encoding during the transmission of the string.

11

Which type of IPAddress is used to represent that there is no network activity



Any

Broadcast

Loopback

None

None of the answers is correct

12

Example of adapter classes are : (Choose all that apply)

BinaryReader

BinaryWriter

StreamReader

StreamWriter

StringReader

StringWriter

13

During Network Programming we are writing applications on which layer (commonly) in the OSI stack 

Network

Application

Session

Presentation

Datalink

Physical

Transport

14

The “using (type variable = initializer)” can be used in any “type” 

True

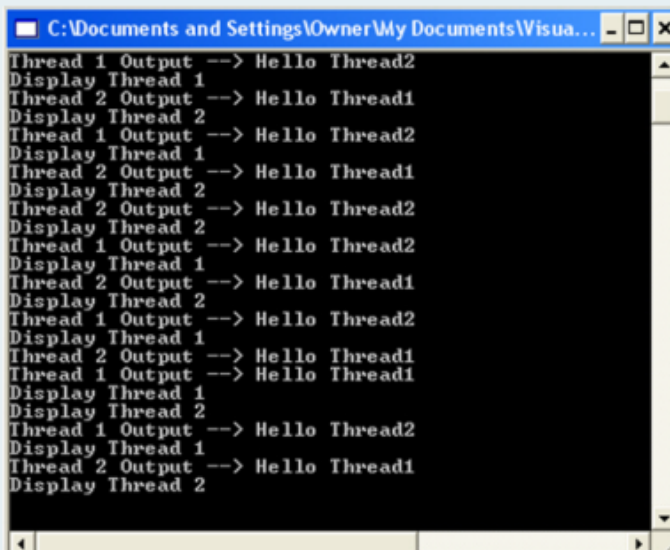
False

15

This is the program output of two threads concurrently accessing one string variable

what is the problem? 

- a. No problem
- b. two threads sharing a common variable in memory
- c. Thread#1 assigns and thread#2 displays !!
- d. Thread#2 assigns and thread#1 displays !!
- e. Both C and D



```

C:\Documents and Settings\Owner\My Documents\Visua...
Thread 1 Output --> Hello Thread2
Display Thread 1
Thread 2 Output --> Hello Thread1
Display Thread 2
Thread 1 Output --> Hello Thread2
Display Thread 1
Thread 2 Output --> Hello Thread1
Display Thread 2
Thread 2 Output --> Hello Thread2
Display Thread 2
Thread 1 Output --> Hello Thread2
Display Thread 1
Thread 2 Output --> Hello Thread1
Display Thread 1
Thread 2 Output --> Hello Thread1
Display Thread 2
Thread 1 Output --> Hello Thread2
Display Thread 1
Thread 2 Output --> Hello Thread1
Thread 1 Output --> Hello Thread1
Display Thread 1
Display Thread 2
Thread 1 Output --> Hello Thread2
Display Thread 1
Thread 2 Output --> Hello Thread1
Display Thread 2

```

f. The code theoretically is executing the two methods DisplayThread1 and DisplayThread2, simultaneously.

16

Common Internet/network applications are :  (Choose all that apply)

Email

File transfer

File sharing

Resource distribution (DNS)

World wide web

Video conference

Gaming

17

For those attributes which you don't want to serialize put _____ on them.



[NonSerializing]

[NonSerialized]

[NonSerial]

[Non]

18

During writing network application on our course you have to be aware of the communication type either wireless or wired. 

Yes

No

19

The main difference between Peek() and Read() in Stream reader is that:

(Choose all that apply)



Peek puts back the data in the stream

Read removes the data from the stream

Read don't seek the get_pointer

None of the answers are correct

20

The order Create/bind/RecvFrom/sendto/close belongs to

Udp socket

Udp server

Udp client

None of the answers is correct

Option 2

21

What did the following pic represent (according to our lecture)



Concurrency

Parallel programming

Serial programming

Multithreading



22

What did The socket define?  (Choose all that apply)

A specific communication domain

stream or datagram

TCP or UDP

A specific protocol

23

There is a fair amount of overhead involved with establishing the connection, but once it is established, data can be reliably transferred between the devices.



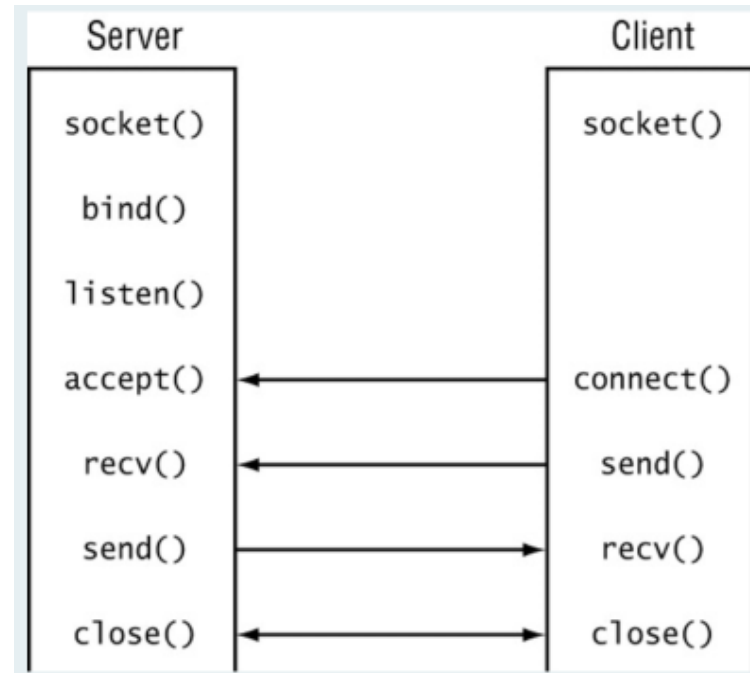
Connection-Oriented Sockets

Connectionless Sockets

24

1. During our course we are aiming to write applications to run over: 

- a. Circuit switched networks
- b. Packet switched networks
- c. Frame relay networks
- d. Asynchronous transfer mode (ATM) networks



25

This Figure, there is no bind and listen in the client why? 

The client uses a pre-established connection and does not need to bind to a specific address, and listen is used only for managing server-side connections.

The client can only send requests on established sockets, and binding or listening is automatically handled by the operating system for outbound connections

The client requires a static IP address for binding and does not need to listen since the server handles incoming requests.

Binding is unnecessary because the client uses a dynamic port for outgoing connections, while listen is only needed for secure protocols such as HTTPS.

26

The CLR main function is to convert MSIL code into Win32 Native dependent code. ☐

True

False

27

Identify the correct statements ☐ (Choose all that apply)

In socket-based network programming, you do not directly access the network interface device to send and receive packets.

An intermediary file descriptor is created to handle the programming interface to the network.

operating system handles the details of determining which network interface device will be used to send out the data and how.

Socket identify communication domain , type , and protocol only.



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```
Microsoft Visual Studio Debug Console  
Original string: A æ u å æ o i æ å  
String via UTF-8: A æ u å æ o i æ å  
ASCII converted string: A ? u ? ? ? i ?  
C:\Users\Dr. Ali\source\repos\WindowsFo  
Press any key to close this window . .
```

The ASCII encoding can only represent characters from the basic Latin alphabet, and characters like "æ", "å", and "ø" are outside its representable range. ✓

The question marks are a result of improper conversion between encodings, where the original non-ASCII characters are lost during conversion. ✓

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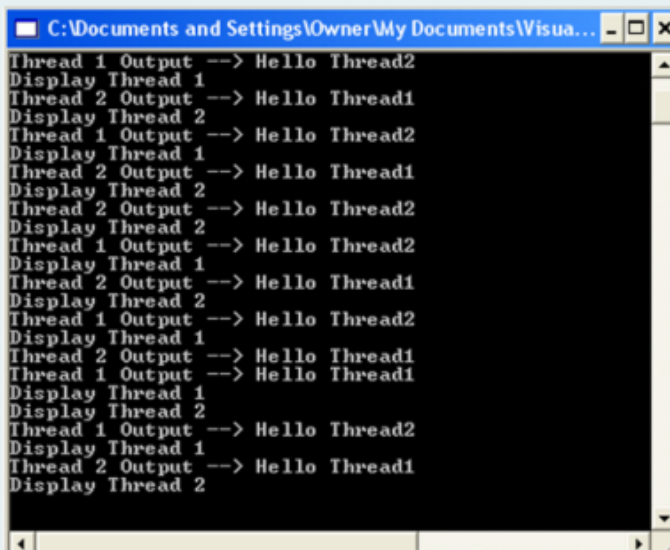


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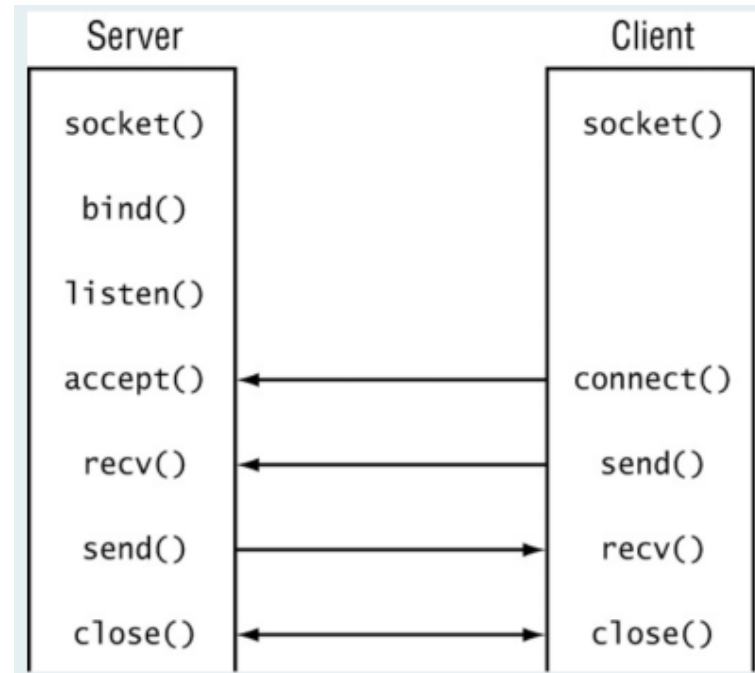


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